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PCB Defect Detection Based on Improved YOLOv8n

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***Abstract*—To address potential defects caused by processing errors during PCB production and ensure both manufacturing efficiency and yield rates, this study introduces an improved defect detection approach based on YOLOv8n. Firstly, feature extraction capability is enhanced by integrating a High Frequency Prior Module (HFPM) and Decomposed Kernel Attention Module (DKAM) into the backbone network. Secondly, multi-scale feature integration is optimized through a Dynamic Feature Fusion Network (DFFN). Thirdly, a Depthwise Separable Decoupled Head (DSDH) is used to streamline the detection head structure. Finally, measurement precision is improved through a loss function based on NWD. Experiments demonstrate that the improved model features a 1.2% increase in mAP@0.5, a 25.4% reduction on Parameters and a 42% decrease on the Floating Point Operations (FLOPs) compared to the original version, and the frame rate reaches 85 frames per second with 4.57MB model size. The improved algorithm possesses a smaller model size and better detection capability, providing an approach for PCB quality inspection in industrial production and deployment on embedded systems.**

Keywords—PCB detection, YOLOv8n, lightweight, deep learning, object detection

I. Introduction

Conventional PCB defect detection techniques (e.g., manual visual inspection, electrical testing, infrared inspection [1]) exhibit critical shortcomings, including low accuracy, frequent missed detections, and false positives.

Image processing techniques can simplify the complex information in PCB images, highlighting key areas for inspection. Replacing traditional image recognition with deep learning further enhances the accuracy and efficiency of defect identification and classification [2]. Representative deep learning object detection algorithms, including CNN, Faster R-CNN, SSD, and YOLO [3-6], fail to optimize the trade-off among detection accuracy, inference speed, and model complexity. In previous studies, the incorporation of high-frequency priors has been neglected when focusing on global information [7]. And convolutional kernel attention with large depth causes high computational and memory costs, reducing model efficiency as kernel size increases [8].

To address these problems, an enhanced YOLOv8n model is proposed in this study. The main contributions include:

1.The backbone is modified with the original C2f module and SPPF module replaced with a High-Frequency Prior Module (HFPM) and Decomposed Kernel Attention Module (DKAM) respectively to reduce computational overhead. 2.The Neck is improved by using a Dynamic Feature Fusion Network (DFFN) which enhances multi-scale feature integration. 3.The detection head is redesigned with feature channel compression and asymmetric branch architecture, introducing a Depthwise Separable Decoupled Head (DSDH) to reduce computation while preserving accuracy. 4.A new loss function is adopted to improve detection precision, replacing the original IoU metric with the normalized Wasserstein distance (NWD).

Experiments show that the proposed improvements are effective and feasible, achieving higher detection accuracy while boosting speed and efficiency, suitable for computation-constrained environments. Fig.1 illustrates the modified network's structural design, where the bold red borders highlight the modified modules.

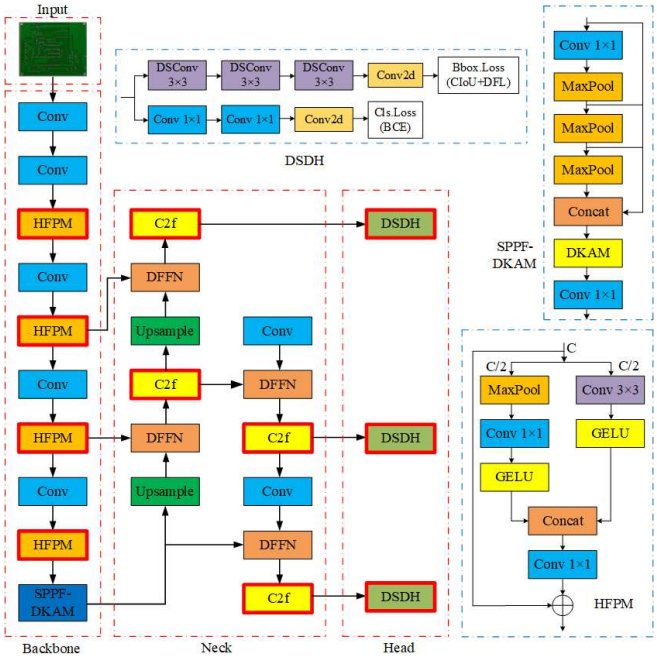


Fig. 1. Structure of the improved model.

II. Method For This Paper

A. Improved Backbone

In the backbone network, the original C2f module and SPPF module are replaced with the High-Frequency Prior Module (HFPM) and Decomposed Kernel Attention Module (DKAM) respectively to reduces redundant computation while improving detail representation. HFPM, illustrated in Fig.2, enhances high-frequency information through two primary branches. One of the two branches extracts local high-frequency features through an architecture containing a 3×3 convolutional layer and a GELU activation. The other branch is high-frequency enhancement branch, focusing on complex high-frequency components (e.g., edges and corners) to capture finer details. By splitting input features into two parts, the module reduces redundant computation while improving detail representation.

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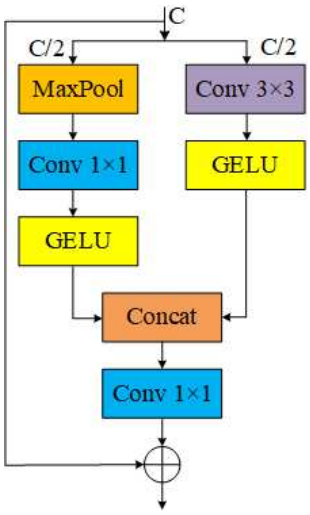
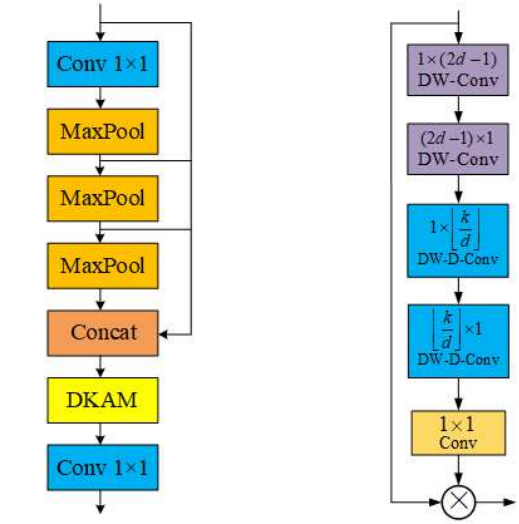


Fig. 2. High-Frequency Prior Module.

Utilizing large kernel attetion modules in the vision attention network leads to superior performance while causing high computational and memory costs. To avoid reducing model efficiency as kernel size increases, the DKAM is proposed to replace the original design of large-depth convolutional kernels. As shown in Fig.3(a), DKAM is integrated into the SPPF module to foucus on the incorporation of high-frequency priors. It breaks down the 2D convolutional kernel of a depthwise layer into cascaded horizontal and vertical 1D kernels, as shown in Fig.3(b). this decomposition enables the use of large-kernel depthwise convolutions within attention modules without requiring additional blocks.



(a)Improved SPPF module (b)DKAM

Fig. 3. Decomposed Kernel Attention Module.

The output of DKAM can be obtained as follows:

$$\overline{Z}^C = \sum_{H,W} W^C_{(2d-1)\times 1} * \left(\sum_{H,W} W^C_{1\times(2d-1)} * F^C\right) \quad (1)$$

$$Z^C = \sum_{H,W} W^C_{\lfloor \frac{k}{d} \rfloor \times 1} * \left(\sum_{H,W} W^C_{1\times \lfloor \frac{k}{d} \rfloor} * \overline{Z}^C\right) \quad (2)$$

$$A^C = W_{1\times 1} * Z^C \quad (3)$$

$$\overline{F}^C = A^C \otimes F^C \quad (4)$$

where d denotes the expansion ratio. $\overline{Z}^C$, Z^C correspond to the depthwise convolution outputs with different kernel sizes. Subsequent experiments demonstrate that this modification reduces computational complexity and memory usage while maintaining accuracy.

B. Dynamic Feature Fusion Network

The Dynamic Feature Fusion Network (DFFN) achieves intelligent multi-scale feature fusion through an adaptive feature weighting mechanism, dynamically optimizing the contribution ratios of features from different sources via learned attention weights. As illustrated in Fig.4 the module first performs dimensional alignment of input features using 1×1 convolutions, followed by channel-wise statistic extraction through global average pooling. A lightweight MLP with bottleneck structure subsequently generates normalized attention weights, with adaptive fusion ultimately accomplished through feature modulation. Through conventional re-weighting or channel-spatial joint attention paradigms, DFFN adopts a dynamic allocation strategy specifically tailored for feature fusion tasks, significantly enhancing multi-scale feature integration while maintaining computational efficiency.

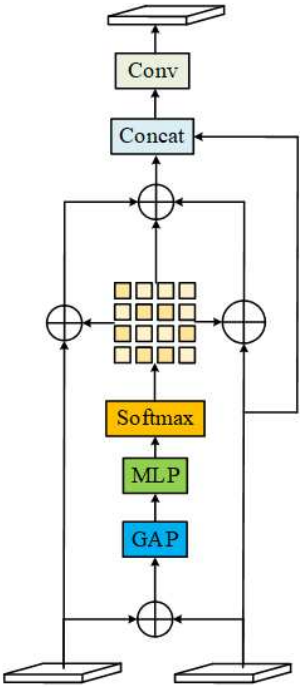


Fig. 4. Dynamic Feature Fusion Network.

C. Depthwise Separable Decoupled Head

The YOLOv8 detector employs a decoupled head structure that significantly enhances detection capability. However, this design increases parameters and cannot fully leverage high-level features extracted by the backbone network, consequently degrading accuracy [9]. Multi-level channel compression is incorporated to develop Depthwise Separable Decoupled Head (DSDH) based on the asymmetric decoupled head. Fig.4 shows our proposed detector head architecture. In both branches, traditional 3×3 convolutions is replaced with 3×3 depthwise separable convolutions (DSConv) [10] and 1×1 convolutions respectively. By substituting the original decoupled head with DSDH in the

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YOLOv8 network, model parameters is decreased and detection speed is improved.

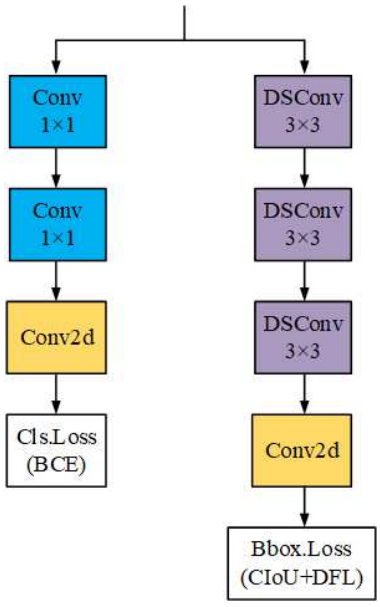


Fig. 5. Depthwise Separable Decoupled Head.

D. Loss Function

The IoU-based metric is highly sensitive to positional deviations of small objects. Degradation of detection performance on small objects is avoided by replacing the original IoU metric with the normalized Wasserstein distance (NWD) embedded in the loss function to calculate distribution distance. As a result, this new metric demonstrates superior performance compared to IoU when measuring the similarity between small objects. For Gaussian distributions and , we define them as:

$$W_2^2(N_a, N_b) = \left\| ([cx_a, cy_a, \frac{w_a}{2}, \frac{h_a}{2}]^T, [cx_b, cy_b, \frac{w_b}{2}, \frac{h_b}{2}]^T) \right\|_2^2 \quad (3)$$

After normalization, we obtain:

$$NWD(N_a, N_b) = \exp\left(-\frac{\sqrt{W_2^2(N_a, N_b)}}{C}\right) \quad (4)$$

where C is a constant closely related to the dataset.

III. Experiments Results And Analysis

A. Experimental details

The experimental platform features a 13th Gen Intel(R) Core(TM) i5-13600KF @3.50 GHz CPU, NVIDIA GeForce RTX 4070 SUPER GPU, and runs on a 64-bit Windows 10 operating system. The development environment is based on Python 3.10.14, PyTorch 2.2.2, and CUDA 12.1.

The PCB dataset in this paper primarily consists of publicly available datasets obtained from online communities, processed through rotation and other augmentations, totaling 5,000 images. It includes six types of defects: open circuit, spurious copper, mouse bite, missing hole, short and spur. The dataset is partitioned into training, validation, and test sets in a 7:2:1 ratio and formatted for YOLO.

B. Ablation experiment

To verify the effectiveness of various proposed improvements to the original algorithm, ablation experiments are conducted. Experiment 1 used the original YOLOv8n algorithm, sequentially incorporating the improved algorithms. A denotes the improved backbone network, including HFPM and SPPF-DKAM; B represents the improved Neck section, featuring the Dynamic Feature Fusion Network (DFFN); C refers to the DSDH detection head; D indicates the NWD loss function. The results of these ablation experiments are presented in Table 1.

TABLE I. Results of Ablation Experiments

Experi-ment	A	B	C	D	Map @0.5%	Para-ms /106	FLOPS /109	FPS
1	×	×	×	×	0.929	3.0	8.1	96
2	√	×	×	×	0.921	2.9	7.2	110
3	×	√	×	×	0.942	3.1	8.5	68
4	×	×	√	×	0.926	2.4	5.7	89
5	×	×	×	√	0.955	3.0	8.1	125
6	√	√	×	×	0.910	2.7	6.7	83
7	√	√	√	×	0.908	2.1	4.3	92
8	√	√	√	√	0.941	2.2	4.7	85

C. Results

Comparative experiments are conducted with SSD, YOLOv5s, YOLOv7-Tiny, YOLOv8n, and YOLOv8m on the same PCB dataset. Metrics include Params, computational complexity (FLOPs), mAP@0.5, model size, and frames per second (FPS). Table 2 presents the training results. The SSD algorithm shows gaps in both accuracy and model parameters compared to other methods. The improved algorithm reduced parameters and FLOPs by 91.47% and 94.02% respectively compared to YOLOv8m, with mAP@0.5 decreased by 3.7%, remaining within acceptable limits. Compared to mainstream lightweight networks YOLOv5s and YOLOv7-Tiny, the improved algorithm achieved parameter reductions of 68.57% and 63.33%, FLOPs reductions of 70.25% and 64.12%, while improving mAP@0.5 by 0.1% and 7.6% respectively. Against YOLOv8n, it reduced parameters, FLOPs, and model size by 25.4%, 42%, and 23.5% respectively, with a 1.2% mAP@0.5 improvement. Experimental results demonstrate that the improved algorithm achieves smaller parameter counts, compact model size, and superior accuracy, enhancing deployment feasibility for PCB detection on low-cost devices and embedded systems.

TABLE II. Comparative Experiments of Different Models

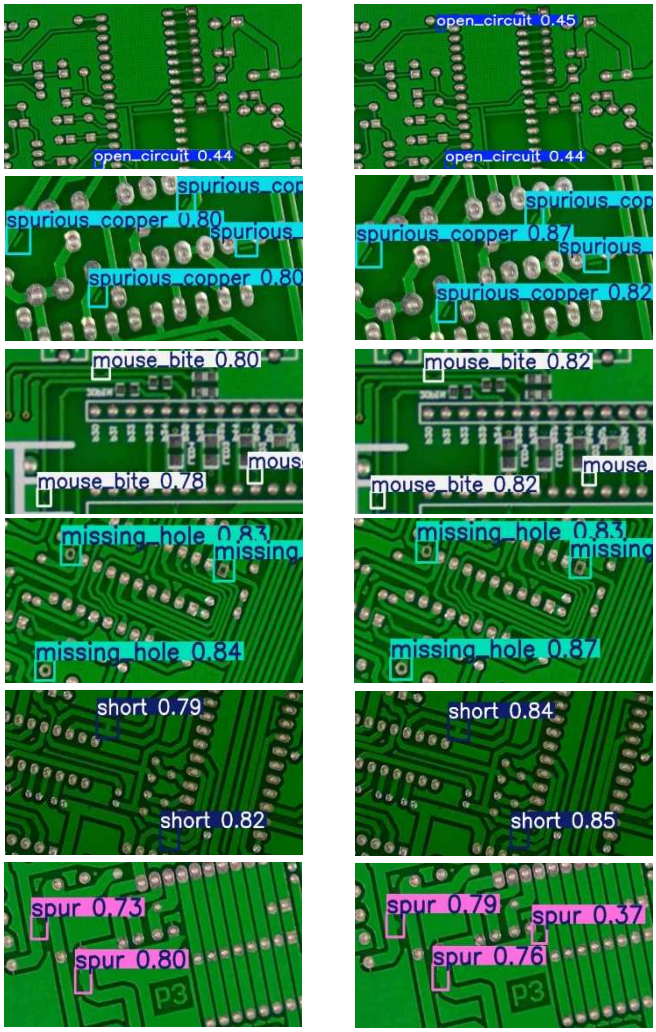
Model	Params /10^6	FLOPs /10^9	mAP @0.5/%	Model size/MB	FPS
SSD	41.3	386	0.790	274	33
YOLOv5s	7.0	15.8	0.940	13.8	78
YOLOv7-tiny	6.0	13.1	0.865	11.7	77
YOLOv8n	3.0	8.1	0.929	5.97	**86**
YOLOv8m	25.8	78.7	**0.978**	49.6	55
ours	**2.2**	**4.7**	0.941	**4.57**	**86**

D. Detection performance of the algorithm

To validate the effectiveness of the improved algorithm for PCB defect detection, comparative tests are conducted between YOLOv8n and our modified model on six defect types: open circuit, spurious copper, mouse bite, missing hole, short, and spur. Fig.5(b) displays outputs of YOLOv8n, while Fig.5(a) shows detection results from our improved model.

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A higher confidence level is observed in the detection of spurious copper, mouse bite, missing hole, and short defects when using the improved algorithm. Correct detection is achieved for open circuit and spur defects, while these instances are missed by the baseline model YOLOv8n.



(a) YOLOv8n (b) ours

Fig. 6. Comparison of detection results.

IV. Conclusion

This study proposes an improved YOLOv8n algorithm tailored for cost-effective and lightweight PCB defect detection in industrial settings, achieving an optimized balance between detection accuracy and inference speed. The HFPM-DKAM module incorporates High-Frequency Prior Enhancement and Decomposed Kernel Attention Module to reduce backbone network parameters while accelerating inference. The Dynamic Feature Fusion Network (DFFN) and Depthwise Separable Decoupled Head (DSDH) maintain detection accuracy while achieving model lightweighting. Further accuracy improvements are attained through modified application of the NWD loss function.

The proposed approach achieves a 25.4% reduction in parameters, 42% decrease in computational load, and 23.5% reduction in model size relative to baseline models, with precision levels preserved. These outcomes sufficiently fulfill industrial specifications for efficiency, accuracy, and computational capacity.

References

[1] ZHAO LY WU YQ, YB YUAN, et al., "Research status and prospect of pcb defect detection algorithm based on machine vision," Chinese Journal of Scientific Instrument, vol. 43, no. 8, pp. 1–17, 2022..

[2] Z. Liu and B. Qu, "Machine vision based online detection of PCB defect," Microprocessors Microsyst., vol. 82, Apr. 2021, Art. no. 103807.

[3] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," Adv. Neural Inf. Process. Syst., vol. 28, 2015.

[4] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "SSD: Single shot multibox detector," 2015.

[5] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), Jun. 2016, pp. 779–788.

[6] C. Ledig, "Photo-realistic single image super-resolution using a generative adversarial network," in Proc. IEEE Int. Conf. Comput. Vis., 2017, pp. 105–114.

[7] K. Nazeri, H. Thasarathan, and M. Ebrahimi, "Edge-informed single image super-resolution," in Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops, 2019, pp. 3275–3284.

[8] M.-H. Guo, C.-Z. Lu, Z.-N. Liu, M.-M. Cheng, and S.-M. Hu, "Visual attention network," Comput. Vis. Media, vol. 9, no. 4, pp. 733–752, 2023.

[9] L. Huang, W. Li, L. Shen, H. Fu, X. Xiao, and S. Xiao, "YOLOCS: Object detection based on dense channel compression for feature spatial solidification," 2023.

[10] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), Jul. 2017, pp. 1800–1807.

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